

BIG BEAR, CA, USA

WMO: 720165

Lat: 34.264N

Lon: 116.854W

Elev: 2057

StdP: 78.94

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-12.5	-10.8	-25.9	0.5	4.5	-22.4	0.6	3.7	9.7	0.6	8.9	0.7	0.2	260	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	27.5	11.8	27.0	11.7	26.0	11.5	14.9	22.1	14.1	22.3	13.5	22.2	3.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	11.6	17.4	11.2	10.6	17.3	9.9	9.8	17.5	48.9	22.4	46.8	21.7	44.5	22.0	17.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.2	-16.9	29.6	1.6	1.1	-18.0	30.4	-19.0	31.0	-19.9	31.6	-21.0	32.4		
(5)			-17.3	16.7	1.6	0.6	-18.5	17.1	-19.4	17.5	-20.3	17.8	-21.4	18.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	0.7	0.4	3.5	6.0	8.6	13.9	16.9	16.3	13.0	8.4	4.1	0.4
	DBStd	6.68	3.59	4.07	3.35	3.28	3.43	2.91	2.10	1.79	2.72	2.66	3.44	3.76
	HDD10.0	1498	289	269	202	122	67	5	0	0	8	61	178	298
	HDD18.3	3886	548	502	460	369	302	134	52	66	161	309	427	556
	CDD10.0	666	0	0	0	3	24	122	214	195	97	11	1	0
	CDD18.3	11	0	0	0	0	0	1	7	2	0	0	0	0
	CDH23.3	1019	0	0	0	0	8	232	427	303	48	1	0	0
	CDH26.7	86	0	0	0	0	0	25	47	14	1	0	0	0
Wind	WSAvg	2.2	2.1	2.4	2.6	2.8	2.5	2.3	1.8	1.7	1.7	1.9	1.9	2.2
Precipitation	PrecAvg	413	104	98	50	26	7	3	7	10	7	20	24	63
	PrecMax	840	448	274	200	74	59	29	33	57	54	149	60	287
	PrecMin	158	8	10	3	5	0	0	0	0	0	2	3	10
	PrecStd	192	129	63	52	18	12	7	9	15	12	32	18	63
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	16.1	17.8	21.2	24.1	29.0	29.8	27.8	26.2	22.4	20.2	16.3
		MCWB	4.2	4.5	6.1	7.7	8.9	11.3	11.7	12.0	10.9	8.0	6.9	4.1
	2%	DB	12.5	14.0	16.2	19.0	22.4	27.3	27.7	27.2	24.9	21.0	17.5	13.5
		MCWB	3.1	4.0	5.0	6.9	8.4	10.7	12.2	11.9	10.8	7.8	5.7	3.7
	5%	DB	11.0	12.2	14.1	17.4	21.0	26.1	27.2	26.2	23.0	19.9	15.9	11.2
		MCWB	2.5	3.0	4.1	5.8	7.8	10.3	12.2	11.8	10.4	7.5	5.1	2.6
	10%	DB	8.0	9.1	12.5	15.9	18.8	24.8	26.1	25.1	22.5	17.8	12.9	7.9
		MCWB	1.2	1.9	3.6	5.0	6.9	9.8	12.0	11.7	10.3	6.4	4.1	1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	5.9	7.0	9.2	10.1	13.5	16.0	15.7	14.8	11.3	7.9	6.7
		MCDB	10.8	11.1	15.6	16.9	21.6	24.1	22.6	22.1	20.0	17.0	16.9	11.1
	2%	WB	4.4	4.5	5.9	7.6	9.1	12.2	15.1	14.8	13.6	9.5	6.6	5.0
		MCDB	9.8	12.3	14.2	17.5	20.7	24.4	22.3	22.2	20.5	16.1	15.4	10.5
	5%	WB	3.4	3.5	5.1	6.5	8.2	11.3	14.4	14.0	12.6	8.3	5.7	3.7
		MCDB	8.3	10.5	13.0	16.3	19.5	24.0	22.5	22.4	20.5	16.9	14.1	9.2
	10%	WB	2.4	2.4	4.1	5.5	7.5	10.4	13.8	13.2	11.4	7.5	4.6	2.5
		MCDB	7.2	8.5	11.4	14.4	17.8	23.0	22.4	22.5	19.8	16.6	12.5	7.7
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.5	13.6	14.7	15.5	18.3	16.8	17.4	17.3	16.7	15.4	12.8
		MCDBR	17.2	18.1	17.9	18.7	19.8	20.6	18.4	18.7	18.5	20.4	19.3	18.1
		MCWBR	9.5	9.9	9.4	9.5	9.9	10.0	8.5	8.7	8.7	10.6	10.6	10.0
	5% WB	MCDBR	13.8	15.9	16.0	17.5	18.3	19.2	14.6	15.3	15.6	17.0	17.2	14.9
		MCWBR	8.3	9.3	9.0	9.4	9.5	9.4	7.1	7.4	7.8	9.3	10.0	8.9
Clear-Sky Solar Irradiance	taub	0.233	0.239	0.248	0.260	0.269	0.271	0.296	0.289	0.273	0.262	0.238	0.237	
	taud	2.648	2.654	2.610	2.574	2.578	2.588	2.559	2.569	2.604	2.603	2.672	2.655	
	Ebn at Noon	1010	1033	1038	1029	1014	1006	979	985	996	989	997	986	
	Edh at Noon	71	79	90	98	100	98	101	98	90	82	69	67	
All-Sky Solar Radiation	RadAvg	3.29	4.25	5.72	7.02	7.84	8.39	7.70	7.23	6.29	4.88	3.67	2.98	
	RadStd	0.23	0.37	0.25	0.24	0.31	0.33	0.36	0.27	0.21	0.24	0.15	0.19	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		+0.62	N/A	-1.65	+0.66	N/A	N/A	-100	+215	+114

Nomenclature: See separate page